

ASSIGNMENT 5

Network Testing Tools Theory

Task (1):

```
Microsoft Windows [Version 10.0.26200.9278]
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C:\Users\DELL>ping www.youtube.com

Pinging www.youtube.com [2001:4860:4827:400::] with 32 bytes of data:
Reply from 2001:4860:4827:400::: time=40ms
Reply from 2001:4860:4827:400::: time=23ms
Reply from 2001:4860:4827:400::: time=14ms
Reply from 2001:4860:4827:400::: time=32ms

Ping statistics for 2001:4860:4827:400:::
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 14ms, Maximum = 40ms, Average = 27ms
```

Task (2):

```
C:\Users\DELL>tracert www.flipkart.com

Tracing route to flipkart.com [103.243.32.90]
over a maximum of 30 hops:

  1      3 ms      2 ms      3 ms      192.168.1.1
  2      7 ms      7 ms      6 ms      10.240.12.32
  3      *         *         *         Request timed out.
  4      5 ms      4 ms      4 ms      125.21.60.117
  5     37 ms     38 ms     36 ms     182.79.141.28
  6     33 ms     33 ms     35 ms     125.19.119.162
  7      *         *         *         Request timed out.
  8      *         *         *         Request timed out.
  9      *         *         *         Request timed out.
 10      *         *         *         Request timed out.
 11     37 ms     34 ms     33 ms     103.243.32.90

Trace complete.
```

Task (3):

```
C:\Users\DELL>nslookup www.zomato.com
Server: UnKnown
Address: fe80::1633:75ff:feed:45b3

Non-authoritative answer:
Name: e70929.dsca.akamaiedge.net
Addresses: 2600:140f:1e00::1737:f4d0
           2600:140f:1e00::1737:f4c3
           23.55.244.120
           23.55.244.138
Aliases: www.zomato.com
          zomato.edgekey.net
```

Task (4):

To test connectivity to another device on your local network:

- Find the device's local IP address, for example **192.168.10.11**.

If the device is online:

Reply from 192.168.10.11: bytes=32
time=2ms TTL=64

If the device is offline:

Request timed out.

Task (5):

1. **Ping** is most useful for checking whether a website or device is reachable; in our test, **ping www.youtube.com** returned replies with an example average response time of 26 ms.
2. **Traceroute** is useful for finding where delays or failures occur along the network path; our **tracert www.flipkart.com** test showed the intermediate network hops.
3. **Nslookup** is best for diagnosing DNS issues by checking whether a domain name resolves to an IP address.
4. In our test, **nslookup www.zomato.com** displayed the IP address associated with Zomato.
5. Therefore, **ping** → **connectivity**, **traceroute** → **network path/slow connection**, and **nslookup** → **DNS problems**.
6. These three tools together provide a quick way to identify common network problems.